

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_diversity_failure_replay_compression.
- Best TSPLIB holdout gap: phase6_diversity_failure_replay_compression.
- Best synthetic holdout gap: phase6_no_replay.

Run Metadata

- run_name: run_20260519_045055_t
- started_at_local: 2026-05-19 04:50:55
- finished_at_local: 2026-05-19 05:11:30
- duration_hhmm: 00:21
- duration_seconds: 1235.548
- seed_offset: 19000
- replicate_label: t
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.093052	0.0	0.059215	0.236772	0.142941	0.0	0.869121	0.76	0.037387
phase6_random_replay	random	score_only	False	0.117651	0.0	0.074869	0.236772	0.158547	0.19457	0.765517	0.76	0.060886
phase6_failure_replay	failure	score_only	False	0.112425	0.0	0.071543	0.171206	0.264522	0.231608	0.0	0.76	0.0

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.119212	0.0	0.075862	0.236772	0.18621	0.19457	0.789905	0.76	0.07546
phase6_stratified_random_replay	stratified_random	score_only	False	0.084157	0.0	0.053554	0.236772	0.159491	0.221162	0.752307	0.76	0.040989
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.153019	0.010343	0.101137	0.236772	0.161314	0.157119	0.0	0.76	0.0
phase6_residual_failure_replay	residual_failure	score_only	False	0.127681	0.010343	0.085013	0.043802	0.148487	0.305	0.0	0.76	0.0
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.103711	0.006169	0.068241	0.171206	0.212967	0.171578	0.829052	0.76	0.036475
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.068295	0.0	0.04346	0.171206	0.230887	0.17767	0.720546	0.76	0.060177

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.093052, synthetic 0.0, combined 0.059215.
- Accepted-epoch count 2, mean accepted code novelty 0.869121, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.142941, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.037387 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.093052 across 7 instances; family means: ch=0.098798, kroD=0.03602, pcb=0.199358, pr=0.078246, rd=0.095702, st=0.044444.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3252, "expected_gap": 0.406514, "family": "a", "name": "a280", "optimality_gap": 0.260954, "residual_gap": -0.14556}, {"best_known_cost": 14379, "cost": 16559, "expected_gap": 0.323861, "family": "lin", "name": "lin105", "optimality_gap": 0.15161, "residual_gap": -0.172251}],

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{"best_known_cost": 7542, "cost": 8109, "expected_gap": 0.244206, "family": "berlin", "name": "berlin52",  
"optimality_gap": 0.075179, "residual_gap": -0.169027}]
```

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.117651, synthetic 0.0, combined 0.074869.
- Accepted-epoch count 2, mean accepted code novelty 0.765517, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.158547, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133845, mean selected residual gap -0.013298.
- Adaptation efficiency 0.060886 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.117651 across 7 instances; family means: ch=0.101298, kroD=0.156147, pcb=0.280436, pr=0.036493, rd=0.125664, st=0.022222.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3497, "expected_gap": 0.406204, "family": "a", "name": "a280", "optimality_gap": 0.355952, "residual_gap": -0.050252}, {"best_known_cost": 7542, "cost": 9789, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52", "optimality_gap": 0.297932, "residual_gap": 0.052825}, {"best_known_cost": 629, "cost": 752, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.195548, "residual_gap": -0.060732}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.112425, synthetic 0.0, combined 0.071543.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.264522, size bias 0.120322, and failure concentration 0.231608.
- Replay selection diversity 0.220764, mean selected expected gap 0.317802, mean selected residual gap -0.008931.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.112425 across 7 instances; family means: ch=0.101681, kroD=0.081948, pcb=0.212671, pr=0.125695, rd=0.114412, st=0.048889.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3336, "expected_gap": 0.408608, "family": "a", "name": "a280", "optimality_gap": 0.293525, "residual_gap": -0.115083}, {"best_known_cost": 14379, "cost": 17342, "expected_gap": 0.326838, "family": "lin", "name": "lin105", "optimality_gap": 0.206064, "residual_gap": -0.120774}, {"best_known_cost": 7542, "cost": 8966, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52", "optimality_gap": 0.188809, "residual_gap": -0.056298}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.119212, synthetic 0.0, combined 0.075862.

- Accepted-epoch count 2, mean accepted code novelty 0.789905, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.18621, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132978, mean selected residual gap 0.000441.
- Adaptation efficiency 0.07546 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.119212 across 7 instances; family means: ch=0.112506, kroD=0.078003, pcb=0.238509, pr=0.155003, rd=0.102402, st=0.035556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3443, "expected_gap": 0.409771, "family": "a", "name": "a280", "optimality_gap": 0.335014, "residual_gap": -0.074757}, {"best_known_cost": 7542, "cost": 9893, "expected_gap": 0.245028, "family": "berlin", "name": "berlin52", "optimality_gap": 0.311721, "residual_gap": 0.066693}, {"best_known_cost": 14379, "cost": 17673, "expected_gap": 0.326977, "family": "lin", "name": "lin105", "optimality_gap": 0.229084, "residual_gap": -0.097893}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.084157, synthetic 0.0, combined 0.053554.
- Accepted-epoch count 3, mean accepted code novelty 0.752307, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.159491, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.237004, mean selected residual gap -0.041653.
- Adaptation efficiency 0.040989 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.084157 across 7 instances; family means: ch=0.066447, kroD=0.170236, pcb=0.143251, pr=0.037907, rd=0.036662, st=0.068148.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 18645, "expected_gap": 0.32496, "family": "lin", "name": "lin105", "optimality_gap": 0.296683, "residual_gap": -0.028277}, {"best_known_cost": 2579, "cost": 3327, "expected_gap": 0.404343, "family": "a", "name": "a280", "optimality_gap": 0.290035, "residual_gap": -0.114308}, {"best_known_cost": 7542, "cost": 8000, "expected_gap": 0.240918, "family": "berlin", "name": "berlin52", "optimality_gap": 0.060727, "residual_gap": -0.180191}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.153019, synthetic 0.010343, combined 0.101137.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.161314, size bias 8.3e-05, and failure concentration 0.157119.
- Replay selection diversity 0.263201, mean selected expected gap 0.155022, mean selected residual gap -0.005724.

- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.153019 across 7 instances; family means: ch=0.105756, kroD=0.170048, pcb=0.223364, pr=0.261079, rd=0.102908, st=0.102222.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3564, "expected_gap": 0.40318, "family": "a", "name": "a280", "optimality_gap": 0.381931, "residual_gap": -0.021249}, {"best_known_cost": 629, "cost": 769, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.222576, "residual_gap": -0.031478}, {"best_known_cost": 7542, "cost": 9205, "expected_gap": 0.236622, "family": "berlin", "name": "berlin52", "optimality_gap": 0.220499, "residual_gap": -0.016123}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.127681, synthetic 0.010343, combined 0.085013.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.043802, mean hardness 0.148487, size bias 0.022181, and failure concentration 0.305.
- Replay selection diversity 0.058403, mean selected expected gap 0.239765, mean selected residual gap 0.051119.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.127681 across 7 instances; family means: ch=0.09147, kroD=0.184794, pcb=0.213951, pr=0.127488, rd=0.095702, st=0.088889.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3184, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.234587, "residual_gap": -0.16836}, {"best_known_cost": 7542, "cost": 8089, "expected_gap": 0.233148, "family": "berlin", "name": "berlin52", "optimality_gap": 0.072527, "residual_gap": -0.160621}, {"best_known_cost": 629, "cost": 660, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.049285, "residual_gap": -0.212718}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.103711, synthetic 0.006169, combined 0.068241.
- Accepted-epoch count 3, mean accepted code novelty 0.829052, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.212967, size bias 0.120322, and failure concentration 0.171578.
- Replay selection diversity 0.196559, mean selected expected gap 0.306518, mean selected residual gap -0.091756.
- Adaptation efficiency 0.036475 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.103711 across 7 instances; family means: ch=0.084267, kroD=0.20541, pcb=0.177222, pr=0.088869, rd=0.063717, st=0.022222.
- Panel synthetic_holdout mean gap 0.006169 across 4 instances; family means: clustered_gaussian=0.024674, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 629, "cost": 875, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.124006}, {"best_known_cost": 7542, "cost": 9644, "expected_gap": 0.237364, "family": "berlin", "name": "berlin52", "optimality_gap": 0.278706, "residual_gap": 0.041342}, {"best_known_cost": 2579, "cost": 3292, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.276464, "residual_gap": -0.125707}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.068295, synthetic 0.0, combined 0.04346.
- Accepted-epoch count 3, mean accepted code novelty 0.720546, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.230887, size bias 0.120322, and failure concentration 0.17767.
- Replay selection diversity 0.198162, mean selected expected gap 0.311259, mean selected residual gap -0.067191.
- Adaptation efficiency 0.060177 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.068295 across 7 instances; family means: ch=0.064705, kroD=0.054804, pcb=0.134743, pr=0.048928, rd=0.076106, st=0.034074.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3480, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.34936, "residual_gap": -0.052811}, {"best_known_cost": 14379, "cost": 18335, "expected_gap": 0.322373, "family": "lin", "name": "lin105", "optimality_gap": 0.275123, "residual_gap": -0.04725}, {"best_known_cost": 629, "cost": 764, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.214626, "residual_gap": -0.049285}]

Judge Appendix

TSP Benchmark Suite Replay-Aware Condition Review

Summary of Best Conditions (Based on Transfer and Optimality Gap)

Metric	Best Condition
Held-out TSPLIB Gap	phase6_diversity_failure_replay_compression (0.0683 final tsplib gap)
Synthetic Holdout Gap	phase6_no_replay (0.0 final synthetic gap)
Transfer (Overall)	phase6_diversity_failure_replay_compression (0.0435 transfer gap)

Detailed Analysis

1. ****Baseline: phase6_no_replay****

- ****Held-out TSPLIB gap:**** 0.0931

- **Synthetic holdout gap:** 0.0 (perfect on synthetic)
- **Transfer gap:** 0.0592
- **Replay mode:** none
- **Archive diversity:** 0.2402 (high)
- **Archive hardness:** 0.1818 (moderate)
- **Replay failure concentration:** 0.0 (no replay)
- **Code novelty:** high (0.869)
- **Interpretation:**
- No replay yields zero gap on synthetic holdouts but moderate transfer gap.
- High code novelty with no replay suggests exploration but less transfer improvement.

2. **Random Replay: phase6_random_replay**

- **Held-out TSPLIB gap:** 0.1177 (worse than no replay)
- **Synthetic holdout gap:** 0.0
- **Transfer gap:** 0.0749 (worse)
- **Archive diversity:** 0.2402 (unchanged)
- **Archive hardness:** 0.1809 (moderate)
- **Replay failure concentration:** 0.28 (moderate concentration)
- **Replay selection diversity:** 0.26 (higher)
- **Code novelty:** 0.766 (decreased from no replay)
- **Interpretation:**
- Random replay slightly degrades transfer and held-out performance.
- Increased failure concentration and replay diversity do not translate to better transfer.
- Code novelty decreases despite replay.

3. **Failure Replay: phase6_failure_replay**

- **Held-out TSPLIB gap:** 0.1124
- **Synthetic holdout gap:** 0.0
- **Transfer gap:** 0.0715
- **Archive diversity:** 0.1747 (lower than baseline)
- **Archive hardness:** 0.3005 (higher hardness)
- **Replay failure concentration:** 0.28 (high)
- **Code novelty:** 0.0 (drops to zero)
- **Interpretation:**
- Failure replay increases hardness but decreases diversity.

- Code novelty drops to zero while transfer gap slightly worsens compared to no replay.
- Replay focuses on failures but at cost of diversity and novelty; transfer not improved.

4. ****Random Replay with Compression: phase6_random_replay_compression****

- ****Held-out TSPLIB gap:**** 0.1192
- ****Synthetic holdout gap:**** 0.0
- ****Transfer gap:**** 0.0759
- ****Archive diversity:**** 0.2402
- ****Archive hardness:**** 0.193 (moderate)
- ****Replay failure concentration:**** 0.16 (moderate)
- ****Replay selection diversity:**** 0.26 (high)
- ****Code novelty:**** 0.79 (moderate)
- ****Interpretation:****
- Compression does not improve transfer or held-out gaps over plain random replay.
- Archive diversity is preserved.
- Code novelty decreased relative to no replay.

5. ****Stratified Random Replay: phase6_stratified_random_replay****

- ****Held-out TSPLIB gap:**** 0.0842 (better than randomized and failure replay)
- ****Synthetic holdout gap:**** 0.0
- ****Transfer gap:**** 0.0536 (second best transfer gap)
- ****Archive diversity:**** 0.2402
- ****Archive hardness:**** 0.1853 (moderate hardness)
- ****Replay failure concentration:**** 0.22
- ****Replay selection diversity:**** 0.19
- ****Code novelty:**** 0.83 (moderate-high)
- ****Interpretation:****
- Stratified random replay improves transfer and held-out performance closer to no replay.
- Replay failure concentration and selection diversity moderate.
- Code novelty is preserved.

6. ****Diversity Weighted Replay: phase6_diversity_weighted_replay****

- ****Held-out TSPLIB gap:**** 0.1530 (worst held-out TSPLIB gap)
- ****Synthetic holdout gap:**** 0.0103 (non-zero)

- **Transfer gap:** 0.1011 (worst transfer gap)
- **Archive diversity:** 0.2402
- **Archive hardness:** 0.1895 (moderate)
- **Replay failure concentration:** 0.44 (high, the highest)
- **Replay selection diversity:** 0.26
- **Code novelty:** 0.0 (drops to zero)
- **Interpretation:**
- Diversity-weighted replay increases replay failure concentration significantly.
- Associated with worst transfer and heldout performance.
- Drops code novelty to zero with increased failure focus.
- Compression-aware replay variant improves this.

7. **Residual Failure Replay: phase6_residual_failure_replay**

- **Held-out TSPLIB gap:** 0.1277
- **Synthetic holdout gap:** 0.0103
- **Transfer gap:** 0.0850
- **Archive diversity:** 0.1752 (low)
- **Archive hardness:** 0.3152 (highest hardness)
- **Replay failure concentration:** 0.31 (high)
- **Replay selection diversity:** 0.06 (low)
- **Code novelty:** 0.0
- **Interpretation:**
- Residual failure replay increases failure concentration and hardness.
- Code novelty is zero.
- Transfer worse than no replay and phase6_diversity_failure_replay_compression.
- Replay selection diversity very low.

8. **Diversity Failure Replay: phase6_diversity_failure_replay**

- **Held-out TSPLIB gap:** 0.1037
- **Synthetic holdout gap:** 0.0062
- **Transfer gap:** 0.0682
- **Archive diversity:** 0.1747 (low)
- **Archive hardness:** 0.2882 (high)
- **Replay failure concentration:** 0.17
- **Replay selection diversity:** 0.20

- **Code novelty:** 0.83 (high)
- **Interpretation:**
- Diversity failure replay raises failure concentration moderately; code novelty high.
- Transfer and held-out gaps better than failure replay alone.
- Archive diversity lower than no replay but better transfer.

9. **Diversity Failure Replay with Compression: phase6_diversity_failure_replay_compression (Best Overall)**

- **Held-out TSPLIB gap:** 0.0683 (best)
- **Synthetic holdout gap:** 0.0 (best)
- **Transfer gap:** 0.0435 (best)
- **Archive diversity:** 0.1747 (low)
- **Archive hardness:** 0.2740 (high)
- **Replay failure concentration:** 0.18 (moderate)
- **Replay selection diversity:** 0.20
- **Code novelty:** 0.76 (moderate)
- **Interpretation:**
- Compression reduces replay failure concentration compared to diversity weighted no compression.
- Best transfer and held-out performance achieved, indicating good generalization.
- Slightly reduced code novelty relative to no replay baseline, despite best transfer.
- Implies compression improves replay quality and transfer without increasing novelty.

Cross-Condition Observations

- **Transfer and Optimality Gap:**
- Best transfer and held-out TSPLIB gaps are achieved by **phase6_diversity_failure_replay_compression**.
- No-replay yields best synthetic holdout gap (zero).
- Replay modes based solely on failure replay or random replay perform worse in transfer and held-out gaps than **diversity_failure_replay_compression**.
- **Replay Failure Concentration:**
- Failure replay and diversity weighted replay (without compression) increase failure concentration and archive hardness.
- Compression reduces failure concentration and improves transfer performance.
- **Archive Diversity and Hardness:**
- No replay and random replay maintain highest archive descriptor diversity (~0.24).

- Failure-focused replay modes tend to lower archive diversity (~0.17) but increase hardness (~0.27-0.31).
- **Code Novelty:**
- Drops to zero when failure or diversity weighted replay without compression is used.
- Compression and stratified random preserve moderate code novelty (~0.75-0.83).
- No replay condition has highest code novelty (0.87).
- **Mean Transfer Probe Gap Last:**
- Matches overall transfer gap trends; lowest for phase6_diversity_failure_replay_compression (~0.044).

Conclusion

- The **phase6_diversity_failure_replay_compression** condition is most effective by primary metrics:
- It achieves the lowest held-out TSPLIB gap (0.0683) and lowest transfer gap (0.0435).
- It balances failure replay with diversity and compression to reduce failure concentration while maintaining archive quality.
- Code novelty decreases compared to no replay but transfer is improved, indicating effective mechanism rather than mere novelty boost.
- Pure **no-replay** maximizes synthetic holdout performance but has higher transfer gaps.
- Failure replay alone or diversity weighted replay without compression can increase hardness and replay failure concentration but does not improve transfer.
- Compression-aware replay plays a critical role in achieving better transfer despite moderate code novelty.

Summary Table for Key Metrics (Rounded)

Condition	Transfer Gap	Heldout TSPLIB Gap	Synthetic Gap	Replay Failure Concentration	Archive Diversity	Archive Hardness	Code Novelty	Replay Mode	Selection Mode
phase6_no_replay	0.0592	0.0931	0.0	0	0.24	0.18	0.87	none	score_only
phase6_random_replay	0.0749	0.1177	0.0	0.28	0.24	0.18	0.77	random	score_only
phase6_failure_replay	0.0715	0.1124	0.0	0.28	0.17	0.30	0	failure	score_only
phase6_random_replay_compression	0.0759	0.1192	0.0	0.16	0.24	0.19	0.79	random	novelty_gate
phase6_stratified_random_replay	0.0536	0.0842	0.0	0.22	0.24	0.19	0.83	stratified_random	score_only

Condition	Transfer Gap	Holdout TSPLIB Gap	Synthetic Gap	Replay Failure Concentration	Archive Diversity	Archive Hardness	Code Novelty	Replay Mode	Selection Mode
phase6_diversity_weighted_replay	0.1011	0.1530	0.01	0.44	0.24	0.19	0	diversity_weighted	score_only
phase6_residual_failure_replay	0.0850	0.1277	0.01	0.31	0.18	0.32	0	residual_failure	score_only
phase6_diversity_failure_replay	0.0682	0.1037	0.006	0.17	0.17	0.29	0.83	diversity_failure	score_only
phase6_diversity_failure_replay_compression	**0.0435*	**0.0683**	0.0	0.18	0.17	0.27	0.76	diversity_failure_compression	score_only

Key Takeaways for Mechanism Studies

- **Replay archive diversity** is highest without failure replay; failure or residual replays reduce descriptor diversity.
- **Hardness** is highest for failure replay and residual failure replay, indicating replay of harder failure cases.
- **Replay failure concentration** correlates negatively with transfer; lowest concentration in best transfer condition with compression.
- **Failure concentration and replay diversity**: Compression reduces failure concentration and preserves replay selection diversity.
- **Code novelty** drops with failure replay modes but transfer improves with replay compression, suggesting code novelty is not the sole driver.
- **Failure replay variants without compression** are suboptimal for transfer despite capturing hard instances.
- **Stratified random replay** is a viable compromise offering moderate transfer improvement with preserved code novelty.

Summary Interpretation

- **Compression-aware diversity failure replay** most effectively balances replay diversity, failure focus, and transfer performance. Despite slight sacrifice in code novelty, it yields best real-world (TSPLIB) and synthetic transfer metrics.
- Pure no replay yields perfect synthetic holdout but weaker transfer.
- Failure-based replay without compression shows increased failure concentration, lowered diversity and novelty, with no transfer gains.
- Random replay is insufficient to improve transfer reliably.
- Replay mechanisms need careful design to optimize transfer, balancing failure focus, diversity, and compression pressure.

Recommended Focus

- Prioritize methods like **phase6_diversity_failure_replay_compression** for best transfer generalization.
- Avoid failure replay without compression due to negative impact on novelty and transfer.
- Further analyze replay archive evolution and instance difficulty concentration to refine replay selection.
- Investigate compression role in maintaining replay selection diversity while focusing on failures.